

Prediction of Specific 70-mer Oligonucleotide with Neural Network Model

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Abstract

Motivation: Mammalian genomes are highly complex. To identify the specific 70-mer oligonucleotides of each gene in a mammalian gene database is a computation intensive task. With the advent of parallel genetic analysis methods such as microarrays and the availability of more and more whole genome sequences of organisms, an algorithm allowing speedy prediction of the specific oligonucleotides with a neural network will be a very useful tool.

Result: We present a computational model for specific 70-mer oligonucleotides prediction based on a Multi-Layer Backpropagation neural network using a broad range of input parameters. The algorithm was applied to the assemblies of gene sequences and was highly efficient for the TIGR Rat and *S.cerevisiae* TC database. The confidence can achieve 94.47% with coverage 22.52% in RGI test performance and it can achieve 96.00% with coverage 79.04% in SCGI test performance. Even the confidence of cross-species neural network can achieve 95.67% with coverage 93.49%. On the other hand, the trained neural network process 1,000 70-mer oligos that just costs 1.74 seconds.

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1 Introduction

DNA microarrays could generate false positive data due to cross-hybridization between highly similar sequences (Xu et al., 2001, Evertsz et al., 2001, Modrek and Lee, 2002). Several studies have concluded that a threshold of around 70% sequence similarity can be used as a reference for cross-hybridization prediction (Kane et al., 2000, Xu et al., 2001, Hughes et al., 2001). Literature data (Hughes et al., 2001) indicate that longer oligonucleotides (60-80mer) provide significant sensitivity.

The algorithm described in this paper predicts the 70-mer specific oligos that have less than 70% similarity to the non-target sequences in a short time. The specific oligos can be applied as candidate probes for gene-specific expression analysis with minimal cross-hybridization problem.

A popular machine learning approach that effectively handles noise and complex relationships in a robust way is the artificial neural network (Baldi, 1998). Artificial neural networks have commonly been used as classifiers. Wu et al. have devised a neural network system for the automatic classification of protein sequences according to super-families (C. Wu, 1992). They have also been used to predict antisense oligo (Alistair M. Chalk 2002).

Two of the best-known heuristic algorithms of sequence alignment are FASTA and BLAST. In this paper, we present a data compilation of 70-mer oligos with the determined similarity value by WU-BLAST (Gish, 1996 - 2004), and apply a neural network model to predict oligo specificity. The execution time of BLAST is much longer than the trained neural network execution time.

2 Database

In this paper, we used the assembled Rat tentative consensus (RGI TC release 11.0) database (Quackenbush, 2001) from The TIGR Gene Index Databases and applied the algorithm to drive the neural network training set. We randomly selected 10 TCs from TC41xxxx (Range TC410000 ~ TC419999) for training data source and 100 TCs from TC42xxxx (Range TC420000 ~ TC429999) for test data source. We chose the small training set because the neural network training needs to learn 10,000 epochs.

General training set: We randomly selected 100 “70-mer oligo” at every TC from training data source. So the general training set has 1,000 oligos.

Average training set: We randomly selected 500 low-similarity (less than 70% similarity) and 500 high-similarity (great than 70% similarity) “70-mer oligo” from training data source. The average training set also has 1,000 oligos.

Test set: We randomly selected 100 “70-mer

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oligo” at every TC from test data source. The test set will have 10,000 oligos.

3 Derived parameters as network input

We created 6 ~ 12-mer subsequence frequencies distributions for the whole RGI TCs then used average frequencies for low frequencies threshold and a half of average frequencies for very low frequencies threshold. The average frequencies are presented in Table 1.

Every l -mer subsequence is encoded by the following formula:

$$code = \sum_{i=1,l} c_i \times 4^{i-1}$$

Where c_i is 0, 1, 2, or 3 for A, C, G or T at the i -th base of the sequence segment and l is the length of the subsequence. For example, a sequence such as ACGT has the numerical code of $0 \times 4^0 + 1 \times 4^1 + 2 \times 4^2 + 3 \times 4^3 = 228$, and $l = 4$.

N-mer	Average frequencies	Half average frequencies
6-mer	18691.71	9345.86
7-mer	4661.17	2330.59
8-mer	1162.38	581.19
9-mer	289.88	144.94
10-mer	72.31	36.16
11-mer	18.44	9.22
12-mer	5.52	2.76

Table 1. The average frequencies

A set of parameters was derived from the features contained in the 70-mer-oligo selections, including values for 5 groups:

(1) Shannon’s entropy of the 6-mer and 8-mer subsequence frequencies distributions:

$$H(\cdot) = - \sum f \log_2 f$$

Where $H(\cdot)$ is the Shannon entropy of the relative frequencies of symbol occurrences $\{f\}$ in the corresponding subsequence (P. Bernaola Galvan, 1996; Shannon, 1948).

(2) The number of low frequencies subsequences (less than average frequencies) for 6, 7, 8, 9, 10, 11, and 12-mer. And the number of very low frequencies subsequences (less than half average frequencies) for 6, 7, 8, and 9-mer.

(3) The number of unique subsequences (frequency is equal to 1) for 10, 11, and 12-mer.

(4) The sum of the frequencies of the subsequences for 6, 7, 8, 9, 10, 11, and 12-mer.

(5) The similarity value by WU-BLAST (Gish, 1996-2004). The high sensitivity of similarity is requirement in the training set. We apply the parameters “word length $W = 7$, matching score $M = 1$, mismatching score $N = -1$, no gap, significance threshold $E = 10$ ” to WU-BLAST.

So the training and test set have 23 input parameters and one desired output item (similarity value from WU-BLAST).

4 Neural Network

A simple outline of the operation for neural network training and prediction was shown in Fig 1. We use the Java Object Oriented Neural Network (JOONE, 2004) that is an open source project. JOONE offers a highly adaptable neural network for Java programmers. The JOONE project source code is covered by a Lesser GNU Public License (LGPL).

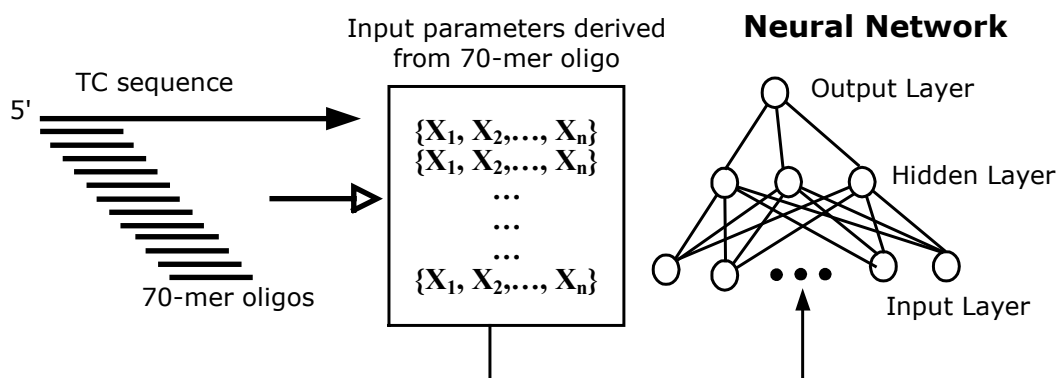


Fig 1. Simple outline of the operation for neural network training and prediction

The training function we used is a batch mode training algorithm. The training is terminated when the number of iterations exceeds the number of epochs (10,000) or when the Root Mean-Square Error (RMSE) value is less than 0.01. We employed the sigmoid layer as both input and output layer. Sigmoid function's output is smoothly limited within the range 0 and 1. The hidden layer is the logarithmic layer. The behavior of logarithmic layer permits to avoid the saturation of the processing elements of a layer in presence of a lot of input synapses connected, or in presence of input values very near to the limits 0 and 1.

Logarithmic activation function:

$$\begin{aligned} y &= \log(1 + x) & \text{if } x \geq 0 \\ y &= \log(1 - x) & \text{if } x < 0 \end{aligned}$$

Sigmoid activation function:

$$y = \frac{1}{1 + e^{-x}}$$

For the current application, the learning rate was set to 0.2, with momentum 0.5 at all simulations. In present study, a backpropagation neural network with one hidden layer was selected. The number of hidden layer nodes is selected based on the classification performance on training data. We trained 4 networks (10, 12, 14 and 16 hidden layer nodes) at all simulation and selected the best performance according to RMSE.

5 Result

The neural network's output value can indicate the specificity of the tested 70-mer oligo. We suppose that an oligo will be the specific oligo (similarity less then 70%) if its score from neural network's output is less than "score threshold". Then we calculated the criterions according to the blast similarities.

We assume that the neural network predicts N specific oligos, M predicted oligos are wrong in the N oligos, and there are P determined specific oligos (similarity less then 70%) by blast. We defined two criterions confidence and coverage to measure the test performance:

$$\text{Confidence} = M/N$$

$$\text{Coverage} = (N-M)/P$$

The results are shown in Table 2, Table 3, Fig 1 and Fig 2. As we can see, the confidence with general training set cannot great than 88% but the confidence with average training set can achieve 94.47% with coverage 22.52%. The average training set is better in this point of view.

Score threshold	Confidence (%)	Coverage (%)
0.80	87.29	96.37
0.81	87.25	96.52
0.82	87.23	96.70
0.83	87.19	96.98
0.84	87.13	97.25
0.85	87.04	97.55
0.86	86.94	97.78
0.87	86.92	98.03
0.88	86.88	98.29
0.89	86.89	98.53
0.90	86.87	98.89
0.91	86.78	99.10
0.92	86.77	99.51
0.93	86.65	99.71
0.94	86.52	99.87

Table 2. Test performance with general training set in a 23-14-1 neural network. Training RMSE = 0.1257.

Score threshold	Confidence (%)	Coverage (%)
0.80	93.40	10.32
0.81	93.40	12.44
0.82	94.06	15.38
0.83	94.38	18.67
0.84	94.47	22.52
0.85	94.15	26.26
0.86	94.16	30.60
0.87	94.06	35.55
0.88	93.68	40.12
0.89	93.48	45.12
0.90	92.82	50.25
0.91	92.03	56.02
0.92	91.46	97.15
0.93	90.75	99.11
0.94	89.96	99.83

Table 3. Test performance with average training set in a 23-16-1 neural network. Training RMSE = 0.1082.

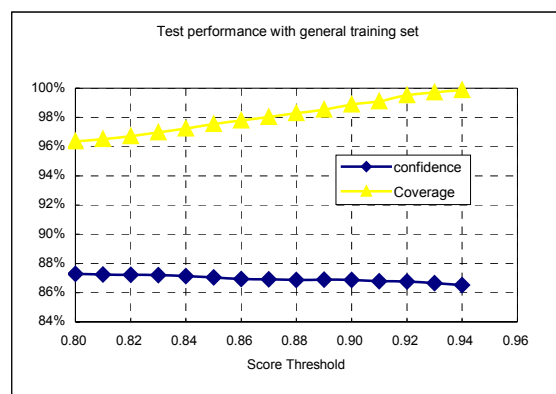


Fig 2. Test performance with general training set

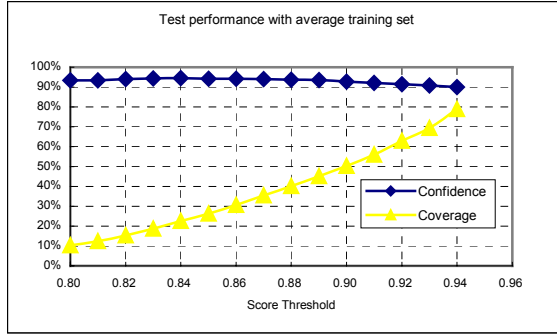


Fig 3. Test performance with average training set

The execution time of blast is much longer than the trained neural network execution time. The neural network may spend long time to discover the elegant network in training stage but it has great efficiency in test stage. The blast execution time of 1,000 oligos is 2,214.76 seconds. The neural network test 1,000 oligos time includes input parameters calculation (25.03 seconds) and process oligos (1.74 seconds). It is 26.77 seconds (1.02% of blast execution time).

6 Discussion

We considered the general training set has low performance because the distribution of randomly selected oligos has a bias. According to the test performance, using average training set can resolve the issue of bias training set and improve the performance.

When considering a multitude of organism sequence databases we asked ourselves the question: “Can the neural network trained by one species apply to another species?” We then used the assembled *S.cerevisiae* tentative consensus (SCGI TC release 3.0) database (Quackenbush, 2001) from The TIGR Gene Index Databases and applied the algorithm to drive SCGI training set and test set by previous methods. The SCGI sequence database has 4,107 TCs and the RGI sequence database has 51,916 TCs. So the SCGI database is related small and it has higher percentage of low similarity oligos than RGI.

To compare the test performance with different species, we use the same test set (SCGI test set) and two different trained neural networks. The results are shown in Table 4 and Table 5. As we can see, the confidence with native network (trained by SCGI) can achieve 96.00% with coverage 79.04% but the confidence with cross-species network can achieve 95.67% with coverage 93.49%. The cross-species network is good enough to perform prediction.

We trained the neural network in 10,000 epochs. It costs long time to try to discover the best network. If we can train neural network once, then

we use it on all of species. It will save much time to train the neural network for every species.

The future directions for the future research might be improvement of training performance by input parameters analysis or study of various N-mer specific oligos prediction.

Score threshold	Confidence (%)	Coverage (%)
0.80	95.95	71.38
0.81	96.00	79.04
0.82	95.97	83.61
0.83	95.96	88.27
0.84	95.91	91.11
0.85	95.89	93.13
0.86	95.89	94.68
0.87	95.79	95.85
0.88	95.67	97.21
0.89	95.41	97.93
0.90	95.34	98.61
0.91	95.29	98.90
0.92	95.22	99.17
0.93	95.17	99.32
0.94	95.03	99.46

Table 4. Test performance of SCGI test set in a SCGI trained 23-12-1 neural network. Training RMSE = 0.0608.

Score threshold	Confidence (%)	Coverage (%)
0.80	95.73	86.37
0.81	95.66	88.06
0.82	95.62	89.71
0.83	95.66	91.50
0.84	95.67	93.49
0.85	95.52	95.01
0.86	95.48	95.90
0.87	95.46	96.61
0.88	95.31	97.36
0.89	95.13	98.10
0.90	94.78	98.54
0.91	94.35	98.82
0.92	93.96	99.05
0.93	93.45	99.13
0.94	92.84	99.63

Table 5. Test performance of SCGI test set by RGI trained 23-16-1 neural network. Training RMSE = 0.1082.

References

- Alistair M. Chalk and Erik L.L. Sonnhammer. (2002) Computational antisense oligo prediction with a neural network model. *Bioinformatics*, 18, 1567-1575.
- Baldi,P.F. and Brunak,S. (1998) *Bioinformatics: The machine learning approach*. MIT Press, Cambridge, MA, pp. 91–104.
- Cuticchia,A.J. and Timberlake,W.E. (1993) Probe choice and analysis package-a set of program to aid in choosing synthetic oligomers for contig mapping. *CABIOS*, 9, 201-203.
- C. Wu, G. Whitson, J. McLarty, A. Ermongkonchai, and T.C. Chang, (1992) *Protein Classification Artificial Neural System*, Protein Science, vol. 1, 1992, pp. 667–677.
- Drmanac,S., Stavropoulos,N.A., Labat,I., Vonau,J., Hauser,B., Soares,M.B. and Drmanac,R. (1996) Gene-representation cDNA clusters defined by hybridization of 57419 clones from infant brain libraries with short oligonucleotide probes. *Genomics*, 37, 29-40.
- Evertsz,E.M., Au-Young,J., Ruvolo,M.V., Lim,A.C. and Reynolds,M.A. (2001) Hybridization cross-reactivity within homologous gene families on glass cDNA microarrays. *Biotechniques*, 31, 1182, 1184, 1186.
- Fu,Y.X., Timberlake,W.E. and Arnold,J. (1992) On the design of genome mapping experiments using short synthetic oligonucleotides. *Biometrics*, 48, 337-359.
- Gish, W. (1996-2004) <http://blast.wustl.edu>.
- Herwig,R., Schmitt,A.O., Steinfath,M., O'Brien,J., Seidel,H., Meier-Ewert,S., Lehrach,H. and Radelof,U. (2000) Information theoretical probe selection for hybridization experiments. *Bioinformatics*, 16, 890-898.
- Hughes,T.R., Mao,M., Jones,A.R., Burchard,J., Marton,M.J., Shannon,K.W., Lefkowitz,S.M., Ziman,M., Schelter,J.M., Meyer,M.R. et al. (2001) Expression profiling using microarrays fabricated by an ink-jet oligonucleotide synthesizer. *Nat. Biotechnol.*, 19, 342-347.
- Java Object Oriented Neural Network (JOONE). (2004) <http://joone.sourceforge.net/>.
- Kane,M.D., Jatke,T.A., Stumpf,C.R., Lu,J., Thomas,J.D. and Madore,S.J. (2000) Assessment of the sensitivity and specificity of oligonucleotide (50mer) microarrays. *Nucleic Acids Res.*, 28, 4552-4557.
- Li,F. and Stormo,G.D. (2001) Selection of optimal DNA oligos for gene expression arrays. *Bioinformatics*, 17, 1067-1076.
- Lockhart,D.J., Dong,H., Byrne,M.C., Follettie,M.T., Gallo,M.V., Chee,M.S., Mittmann,M., Wang,C., Kobayashi,M., Horton,H. and Brown,E.L. (1996) Expression monitoring by hybridization to high-density oligonucleotide array. *Nat. Biotechnol.*, 14, 1675-1680.
- Modrek,B. and Lee,C. (2002) A genomic view of alternative splicing. *Nat. Genet.*, 30, 13-19.
- P. Bernaola Galvan, R. Roman Roldan, J.L. Oliver. (1996) Compositional Segmentations and long-range fractal correlations in DNA Sequences. *Physical Review E* 53, pp 5181-5189.
- Pei-chin Chang and Konan Peck. (2002) Design and assessment of a fast algorithm for identifying specific probes for human and mouse genes. *Bioinformatics*, 19, 1311-1317.
- Quackenbush,J., Cho,J., Lee,D., Liang,F., Holt,I., Karamycheva,S., Parvizi,B., Pertea,G., Sultana,R. and White,J. (2001) The TIGR Gene Indices: analysis of gene transcript sequences in highly sampled eukaryotic species. *Nucleic Acids Res.*, 29, 159-164.
- Religio,A., Schwager,C., Richter,A., Ansorge,W. and Valcarcel,J. (2002) Optimization of oligonucleotide-based DNA microarrays. *Nucleic Acids Res.*, 30, e51.
- Rouillard,J.M., Herbert,C.J. and Zuker,M. (2002) OligoArray : genome-scale oligonucleotide design for microarrays. *Bioinformatics.*, 18, 486-487.
- SantaLucia,J.,Jr. (1998) A unified view of polymer, dumbbell, and oligonucleotide DNA nearest-neighbor thermodynamics. *Proc. Natl. Acad. Sci. USA*, 95, 1460-1465.
- Shannon, C. E. (1948) A mathematical theory of communication. *The Bell System Technical Journal*. 27:379–423; 623–656.
- Snehasis Mukhopadhyay, Changhong Tang, Jeffery Huang And Mathew Palakal. (2003) Genetic sequence classification and its application to cross-species homology detection. *Journal of VLSI Signal Processing* 35, 273–285.
- The TIGR Gene Index Databases, The Institute for Genomic Research, Rockville, MD 20850 (URL: <http://www.tigr.org/tdb/tgi>)
- Tijssen,P. (1993) *Hybridization with nucleic acid probes, Part I: Theory and nucleic acid preparation*. Amsterdam; Elsevier, New York, pp. 19-79.
- Xu,W., Bak,S., Decker,A., Paquette,S.M., Feyereisen,R. and Galbraith,D.W. (2001) Microarray-based analysis of gene expression in very large gene families: the cytochrome P450 gene superfamily of *Arabidopsis thaliana*. *Gene*, 272, 61-74.
- Yuan,Q., Ouyang,S., Liu,J., Suh,B., Cheung,F., Sultana,R., Lee,D., Quackenbush,J. and Buell,C.R. (2003) The TIGR rice genome annotation resource: annotating the rice genome and creating resources for plant biologists. *Nucleic Acids Res.*, 31, 229-233.